**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 037 · Operating Systems: Interfaces and Memory

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. After marking an assessment, name one error category you should identify.

Q2. Which is malware?
A. Ransomware
B. A trusted security update
C. A spreadsheet formula
D. An authorised backup

Q3. Given A has code 65 and capital-letter codes are consecutive, give the code for C and explain how you found it.

Q4. A website needs English, Greek and Chinese text. “Unicode has fewer characters than standard ASCII.” Correct this. Why does Unicode suit this site?

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

Operating system — System software managing hardware/resources and supplying services to programs.

GUI — Graphical User Interface using windows, icons, menus and a pointer.

CLI — Command Line Interface using text commands.

Memory management — Allocation and management of RAM for running programs.

Multitasking — Coordinating the execution of multiple applications.

User interface — Means by which the user interacts with a computer.

RAM — Volatile primary memory holding active programs and data.

### New learning

Why an operating system?: System software coordinates hardware and applications, and provides services such as a user interface, memory management and security. Without it, applications would have to manage many hardware details.

Graphical and command interfaces: A GUI lets users select icons, windows and menus, making common tasks approachable. A CLI accepts typed commands, which can be efficient for an experienced user and automation.

Allocating memory: The OS allocates available RAM to applications, manages which data is kept in RAM and moves data as needed. Paging and segmentation are not required for this specification.

Multitasking: The OS manages running applications and shares processor time; it allocates memory and handles resources so users can switch between tasks. Switching does not mean one CPU core necessarily runs every task at exactly the same instant.

### Worked example

A student has a browser, music app and editor open. Explain how the OS manages their memory and allows multitasking. [4 marks]

1. Identify the three applications competing for resources.

2. The OS allocates space in RAM to applications [1], keeping their required data available [1].

3. The OS manages CPU time among tasks [1], so the student can switch between or use different applications [1].





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Q2. OS allocates RAM to each program [1] so needed data is available [1]. It manages the running tasks and processor time [1], so the user can switch between programs while both remain active [1].

Q3. GUI uses visual windows/icons and menus [1], making routine functions easier to discover [1]. CLI takes typed commands [1] that can be efficient for expert repetitive tasks [1]. OS allocates RAM to programs [1], allowing applications to run with required data in memory [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Missing knowledge, a wrong technical term, a missing cause and effect, or a missing detail.

Q2. A — Ransomware

Q3. 67; C is 2 position(s) after A.

Q4. Unicode supports far more characters than standard ASCII. It can cover English, Greek and Chinese. Use an encoding with enough codes; Unicode characters do not always take one byte.

### I Do / We Do pages · Guided practice

Q1. The OS allocates available RAM to each application [1], allowing program data to be held in primary memory [1]. It manages CPU time or resources [1], letting both applications remain usable through multitasking [1].

Q2. A CLI can run repeated or scripted commands efficiently [1], speeding up frequent technical tasks [1]. Users must remember accurate commands [1], so beginners may find it harder and make errors [1].

Q3. The OS must allocate the available RAM across programs [1] to prevent applications from competing for the same allocated space [1]. It manages moving/loading application data in memory [1], so programs have needed data as the user switches between them [1].

### You Do pages · Code First

```
freeMB = 1200
appMB = 850
print(freeMB - appMB)
```

Expected result: 350

### You Do pages · Independent questions

Q1. Provides a user interface [1]; allocates/manages RAM for applications [1]. Other valid functions include file, device, user management and multitasking.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

4. Four linked marks: memory allocation, purpose, processor time, multitasking.

## We Do · Practise with guidance

### Guided 1

A receptionist runs booking software and an email client together. Explain TWO ways the OS helps these applications operate. [4 marks]

Worked method: The OS allocates available RAM to each application [1], allowing program data to be held in primary memory [1]. It manages CPU time or resources [1], letting both applications remain usable through multitasking [1].

### Guided 2

A technician prefers typed commands for repetitive setup tasks. Explain one benefit and one drawback of a CLI compared with a GUI. [4 marks]

Worked method: A CLI can run repeated or scripted commands efficiently [1], speeding up frequent technical tasks [1]. Users must remember accurate commands [1], so beginners may find it harder and make errors [1].

### Guided 3

A computer has three programs open but limited RAM. Explain TWO memory-management problems the OS must handle. [4 marks]

Worked method: The OS must allocate the available RAM across programs [1] to prevent applications from competing for the same allocated space [1]. It manages moving/loading application data in memory [1], so programs have needed data as the user switches between them [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Predict free RAM after loading the application (MB).

```
freeMB = 1200
appMB = 850
print(freeMB - appMB)
```

predict

### Question 1

State TWO functions of an operating system. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 2

A photographer switches between photo-editing software and a browser. Explain how operating-system memory management and multitasking support this. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 3

Compare GUI and CLI interfaces for an office worker and explain the OS role in memory management. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.



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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Exit Ticket · Answer before checking

The operating system can allocate RAM to different applications to help them run.